C1 Fabric architecture

Workspaces, capacities, items and OneLake as one logical lake.

C1-L1 Tenant, capacity, workspace and item

The capacity decides who can view, how much runs at once and what it costs. Separate environments into workspaces, and give production its own capacity.

Fabric hierarchy	Tenant, capacity, workspace, item.
What F64 means	An F SKU with 64 capacity units. F SKUs run from F2 to F2048.
Free-licence viewers of Power BI content	Need an F64 or larger (or P) capacity. Smaller F SKUs need Pro, PPU or trial licences for viewers.
Pausing a capacity	Stops compute billing from then on, bills remaining smoothed and overage usage at once, and stops its items.

C1-L2 OneLake, one lake per tenant

One lake, one copy, many engines. Address data by GUID in anything automated.

```
https://onelake.dfs.fabric.microsoft.com/<workspace>/<item>.<itemtype>/<path>
abfss://<workspace>@onelake.dfs.fabric.microsoft.com/<item>.<itemtype>/<path>
```

How many OneLakes per tenant?	Exactly one. You cannot create or delete OneLakes.
OneLake table formats	Delta Parquet or Iceberg, open formats every Fabric engine reads.
OneLake path by name	abfss://<workspace>@onelake.dfs.fabric.microsoft.com/<item>.<itemtype>/Tables/<table>.
Why GUIDs in automation	They survive renames, avoid special-character problems in abfs URIs, and drop the item type.

C1-L3 Workspace roles, granted to groups

Roles go to groups, write access in production goes to automation, and people read.

Fabric workspace roles	Admin, Member, Contributor, Viewer.
Admin-only rights	Update or delete the workspace, manage admins, connect Git, create a workspace identity.
Viewer and lakehouse data	Reads with T-SQL (ReadData). Not through Spark or OneLake APIs (ReadAll starts at Contributor).
Who writes to production?	An automation identity as Contributor. People get Viewer, through groups.